Anand Bhattad

Pixels · Perception · Physics

Assistant Professor of Computer Science · DSAI & Institute of Robotics
Johns Hopkins University

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Research

Generative models look right without being right. My 3P Vision group builds the tools to tell the difference — to read out what they know and don't, and how to improve them — along three Ps:

Pixels — controllable generative models for relighting, compositing, and editing.

Perception — reading out the scene structure models implicitly know: intrinsics, geometry, and parts.

Physics — testing and enforcing physical and projective consistency, and carrying it to robots.

The group looks beyond generative models too: vision architectures, 3D perception, multisensory perception for embodied AI, and generative models for computational mechanics.

Keywords: Visual Generative Models; 3D Vision; Vision-based Robotics; Neural Rendering; Inverse Graphics; Intrinsic Images; Image-based Lighting; Physics Awareness in Large Vision Models.

Education

University of Illinois Urbana-Champaign (UIUC), USA

Ph.D., Computer Science 2019 – 2024

Thesis: **Exploring Knowledge in Generative Models**

Thesis committee: **David Forsyth (Advisor)**, Alexei A. Efros, William T. Freeman, Derek Hoiem, Svetlana Lazebnik, Shenlong Wang

M.S., Computer Science 2017 – 2018

M.S., Civil Engineering 2015 – 2017

National Institute of Technology Karnataka (NITK) Surathkal, India

B.Tech., Civil Engineering 2011 – 2015

Work Experience

Johns Hopkins University, Assistant Professor of Computer Science; core member of DSAI & the Institute of Robotics 2025 – present

Toyota Technological Institute at Chicago, Research Assistant Professor 2023 – 2025

University of California Berkeley, Visiting Scholar; Host: Alexei A. Efros 2024

Allen Institute for Artificial Intelligence (Ai2), Research Intern with Aniruddha Kembhavi 2023

Intel, Research Intern with Stephan R. Richter in Vladlen Koltun's team 2021

NVIDIA, Research Intern with Andrew Tao and Bryan Catanzaro 2020

Fyusion Inc, Research Intern with Abhishek Kar and Rodrigo Ortiz Cayon 2019

Siemens Corporate Technology, Computer Vision & DL Intern with Jan Ernst 2017

Michigan State University, Research Intern with Nizar Lajnef 2014

IIIT Hyderabad, Research Intern with Ramacharla Pradeep Kumar 2013

Selected Honors and Awards

Best Paper Award in Robot Learning, ICRA 2026
Outstanding Reviewer, ICCV 2023
Best Paper Finalist, CVPR 2022
Outstanding Emergency Reviewer, CVPR 2021
Excellent Teaching Assistant, Intro Computing: Engrg & Sci 2016

Selected Publications

Self-selected publications (not necessarily the most cited work).

- Generative Cinematographer: Composing Camera and Object Motion in 3D; **on arXiv**
- PowerSim: Differentiable Physics Simulation and Rendering with Power Diagrams; **on arXiv**
- Shadows Don't Lie and Lines Can't Bend! Generative Models Don't Know Projective Geometry...For Now; **CVPR 2024**
- StyleGAN Knows Normals, Depth, Albedo, and More; **NeurIPS 2023**
- PRINCIPIA: Relational Physics Tests for Video Models; **NeurIPS 2026 (Evaluations & Datasets Track)**
- Visual Jenga: Discovering Object Dependencies via Counterfactual Inpainting; **NeurIPS 2025**
- Latent Intrinsics Emerge from Training to Relight; **NeurIPS 2024** (**Spotlight**)
- Generative Blocks World: Moving Things Around in Pictures; **ICLR 2026**
- Do You Know Where Your Camera Is? View-Invariant Policy Learning with Camera Conditioning; **ICRA 2026** (**Best Paper Award in Robot Learning**)
- LumiNet: Latent Intrinsics Meets Diffusion Models for Indoor Scene Relighting; **CVPR 2025**

Advising

Primary or co-advisor at Johns Hopkins University.

Postdoctoral: Jamila Taaki *(primary advisor Laurent Pueyo)*.

Doctoral: Deming (Remus) Li *(co-advised with Rama Chellappa)*, Jiahan Zhang, Trong-Tung Nguyen.

Master's & Undergraduate: Chaohao Yang, Jiafeng Gu, Jin Hong (Jayden) Moon, Rayhaneh Behravesh.

Complete publication list, invited talks, service, and advising at anandbhattad.github.io.